

Matrix Multiplication

Matrix Multiplication via Matrix-Vector Mult

Defn. If matrix A is $m \times n$ and matrix B is $r \times s$, then for the **product** AB to be valid it must be that $n = r$. If valid, the product AB has size $m \times s$. The columns of the product are the results of multiplying the first matrix by the columns of the second.

That is,

$$AB = [Ab_1 \ Ab_2 \ \cdots \ Ab_s]$$

where b_j is the j^{th} column of B .

Example of Matrix Multiplication

The product of a 2×3 and 3×4 matrix is a 2×4 matrix:

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & -2 \end{bmatrix} \begin{bmatrix} 3 & 1 & -1 & 5 \\ -2 & 0 & 3 & -4 \\ 1 & -2 & 2 & -1 \end{bmatrix} = \begin{bmatrix} -2 & 3 & 3 & -2 \\ -8 & 4 & 5 & -10 \end{bmatrix}$$

An example detail: the 3rd column of the result is given by $-\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3\begin{bmatrix} 2 \\ 3 \end{bmatrix} + 2\begin{bmatrix} -1 \\ -2 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$.

Formula for Entry in Product

Note

$$(AB)_{ij} = \sum_k a_{ik} b_{kj}$$

That is, to calculate entry in row i and column j of the product, look at row i of the first matrix and column j of the second matrix; then multiply corresponding entries and add.

$$\begin{bmatrix} \cdot & \cdot & \cdot \\ 0 & 3 & -2 \end{bmatrix} \begin{bmatrix} \cdot & \cdot & -1 & \cdot \\ \cdot & \cdot & 3 & \cdot \\ \cdot & \cdot & 2 & \cdot \end{bmatrix} = \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & 5 & \cdot \end{bmatrix}$$

$(0 \times -1) + (3 \times 3) + (-2 \times 2)$

Matrix Multiplication is Associative

Fact. *Brackets don't matter.*

For example $(AB)C = A(BC)$ (and the one product is valid whenever the other one is).

Matrix multiplication is not Commutative

Fact. *Order matters.*

There is no guarantee that (and it is unlikely that) $AB = BA$.

Indeed, the one product might be valid when the other one is not.

The Identity Matrix

Defn. The **identity matrix** I_n is the $n \times n$ diagonal matrix with 1's on the diagonal. (We sometimes write just I .) Its columns are the vectors e_i : these have 0's in every position except for a 1 in the i^{th} position.

If A is a square matrix, then $IA = AI = A$, where I is the identity matrix of the same size.

Matrix Powers

We use A^p to mean the product of p copies of A .
(This needs A to be square.)

Transpose and Products

Fact. $(AB)^T = B^T A^T$

Note that the order is swapped!

Elementary Row Operations Revisited

Fact. *Each elementary row operation is equivalent to multiplying on the left by a matrix called an **elementary matrix**.*

Summary

If matrix A is $m \times n$ and matrix B is $r \times s$, then the product AB is valid if $n = r$ and has size $m \times s$. Each column of AB results from multiplying A by the column of B . Equivalently $(AB)_{ij} = \sum_k a_{ik}b_{kj}$

Matrix multiplication is associative but not commutative: brackets don't matter but order does.

Summary (cont)

The identity matrix is a diagonal matrix with 1's on the diagonal. Multiplying by the identity leaves a matrix unchanged. We use A^p to mean the product of p copies of A . Each elementary row operation is equivalent to multiplying on the left by an elementary matrix.